

NORTHERN POWER CORP Downward sloping demand curve
Northern Power Corp. is a public utility company that provides electric power to customers in the northern United States. The company's demand curve is downward sloping, indicating that the price of electricity is inversely related to the quantity demanded. The demand curve is represented by the equation $P = 10 - 0.0001Q$, where P is the price per kilowatt-hour and Q is the quantity of electricity demanded in kilowatt-hours. The company's marginal cost curve is represented by the equation $MC = 0.0002Q$, where MC is the marginal cost per kilowatt-hour. The company's marginal revenue curve is represented by the equation $MR = 10 - 0.0002Q$, where MR is the marginal revenue per kilowatt-hour. The company's profit-maximizing output level is determined by setting marginal revenue equal to marginal cost, which yields the equation $10 - 0.0002Q = 0.0002Q$. Solving for Q , we find that the profit-maximizing output level is $Q = 50,000$ kilowatt-hours. The corresponding price is $P = 10 - 0.0001(50,000) = 5$ dollars per kilowatt-hour. The company's total revenue is $TR = P \times Q = 5 \times 50,000 = 250,000$ dollars. The company's total cost is $TC = MC \times Q = 0.0002(50,000) \times 50,000 = 500$ dollars. The company's profit is $\pi = TR - TC = 250,000 - 500 = 249,500$ dollars.